ROAD TRAFFIC CRASH ANALYSIS IN NIGERIA (2021–2023)

Programming Group B7

Gbena Edem Komla (Group leader) – **22425100**

Nana Efua Bo Alan-Mensah – **22424872**

Emmanuel Mensah Quaye - **22428987**

Eric Agbeko Kudah **- 22425198**

Table of Contents

1. Topic Studied 1
2. What Is Known About the Topic 1
3. Why the Topic Is Interesting, Relevant, and Important 1
4. Description of the Data Used 2
5. Tools and Methods Used 2
6. Results and Societal Impact 2
7. Team Members and Responsibilities 3
8. Reflections on the Project 3

Table 1: Names of Team Members showing respective roles 3

1. Topic Studied

This project focuses on the analysis of road traffic crashes in Nigeria between 2021 and 2023. Using national road transport data, the study explores patterns in crash frequency, casualties, contributing factors, vehicle involvement, gender distribution of injuries and deaths, and the geographic spread of crashes across Nigerian states.

2. What Is Known About the Topic

Road traffic accidents are widely recognised as a major public health and development challenge, particularly in low- and middle-income countries. Existing literature shows that developing nations account for a disproportionate share of global road traffic fatalities despite having fewer vehicles per capita. In Nigeria, rapid urbanisation, population growth, and increased vehicle ownership have placed immense pressure on road infrastructure and traffic management systems.

Previous studies consistently highlight human-related factors such as speeding, reckless driving, alcohol consumption, fatigue, and poor compliance with traffic regulations as leading causes of crashes. Vehicle-related issues, including poor maintenance and overloading, as well as infrastructural challenges like bad roads and inadequate signage, further worsen the problem. Research has also shown that males, especially those in the economically productive age group, are more likely to be involved in road traffic crashes, leading to significant social and economic consequences for households and the wider economy.

3. Why the Topic Is Interesting, Relevant, and Important

This topic is particularly interesting because road traffic crashes affect almost everyone either directly or indirectly. Beyond the tragic loss of lives, crashes result in long-term disabilities, reduced productivity, emotional trauma, and significant healthcare costs. In Nigeria, where healthcare systems are already under strain, preventable road traffic injuries place an additional burden on limited resources.

The relevance of this study lies in its practical application. By identifying high-risk states, peak periods, dominant crash causes, and vulnerable groups, the analysis provides evidence that can support targeted road safety campaigns, enforcement strategies, and infrastructure improvements. The project is also timely, as data-driven decision-making is increasingly being adopted by governments and organisations seeking cost-effective solutions to societal challenges. Understanding when, where, and why crashes occur is a critical step toward reducing their frequency and severity.

4. Description of the Data Used

The analysis is based on secondary data obtained from official Nigerian road transport records covering the years 2021 to 2023. Data Source: Federal Road Safety Corps and National Bureau of Statistics https://microdata.nigerianstat.gov.ng/index.php/catalog/164

The dataset includes quarterly records of road traffic crashes across all states, detailing total crash cases and casualties. Additional tables provide information on crash causes, vehicle types and categories involved, gender-based injuries and deaths, and spatial boundary data for Nigerian states.

The data was structured across multiple Excel sheets, each representing a specific aspect of road traffic incidents. While the dataset was comprehensive, it required extensive cleaning, standardisation of state names, handling of missing values, and aggregation across years and quarters to enable meaningful analysis. Despite these challenges, the dataset offered a rich foundation for uncovering trends and patterns in road traffic safety.

5. Tools and Methods Used

Python was used as the primary analytical tool throughout the project due to its flexibility and strong ecosystem for data analysis. The pandas library was employed for data cleaning, transformation, aggregation, and exploratory analysis. Numerical operations were supported using NumPy, while Matplotlib and Plotly were used to create clear and informative visualisations.

Geospatial analysis was conducted using GeoPandas, allowing crash data to be merged with geographic boundary files to produce hotspot maps that highlight high-risk states. In addition, Streamlit was used to develop an interactive dashboard that enables users to explore the data dynamically through filters, charts, and maps. The methods applied include trend analysis, ranking, categorical aggregation, and spatial visualisation, all aimed at making complex data easier to understand and interpret.

6. Results and Societal Impact

The results of the analysis reveal clear patterns in road traffic crashes across Nigeria. States such as FCT with 4,627 crash cases recorded the highest crash volumes followed by Ogun and Nasarawa, identifying them as major hotspots that require focused attention. Crash occurrences varied across quarters and year 2022 recording the highest crashes, indicating seasonal and temporal trends that could be linked to travel behaviour, weather conditions, or economic activity. The total number of casualties across the three years was 126,618.

Human-related factors emerged as the dominant causes of crashes, reinforcing existing findings that driver behaviour plays a critical role in road safety outcomes. Commercial and private vehicles accounted for the majority of crashes, reflecting their high presence on Nigerian roads. Gender-based analysis showed that adult males experienced significantly higher injury and fatality rates than females, highlighting the need for targeted safety education and enforcement.

From a societal perspective, these findings are valuable for policymakers, road safety agencies, urban planners, and public health officials. The insights can guide targeted interventions, such as improved enforcement in hotspot states, public awareness campaigns addressing risky driving behaviour, and infrastructure investments where they are most needed. Ultimately, reducing road traffic crashes contributes to safer communities, lower healthcare costs, and improved economic productivity.

7. Team Members and Roles

Table 1: Names of Team Members showing respective roles

Name	Index No.	Roles
Gbena Edem Komla	22425100	Data integration, profiling and cleaning, Exploratory and descriptive analytics, Spatial visualizations
Nana Efua Bo Alan-Mensah	22424872	Dashboard deployment and final report and presentation
Emmanuel Mensah Quaye	22428987	Assisted with Spatial visualizations
Eric Agbeko Kudah	22425198	Assisted with Data integration, profiling and cleaning

8. Reflections on the Project

This project was a valuable learning experience that strengthened both technical and analytical skills. Working with real-world data highlighted the importance of thorough data cleaning and validation before analysis. The project also demonstrated how visualisation and storytelling can transform complex datasets into insights that are accessible to both technical and non-technical audiences.

Beyond technical skills, the project reinforced the role of data analysis in addressing real societal problems. Road traffic crashes are not just numbers in a dataset; they represent lives affected and opportunities lost. This work has deepened our appreciation for evidence-based decision-making, ultimately showcasing the transformative power of data science to catalyze public safety initiatives and bolster national development.